BSR PROJECT

"Exploring Cancer Trends in India: A Statistical Approach to Gender and Zonal variation"

Department of Statistics

Patna Women's College

(Autonomous)

Patna University

4th Cycle NAAC Accredited at 'A++' grade

"College with Potential of Excellence" (CPE)

Status accorded by UGC

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PATNA WOMEN'S COLLEGE

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CERTIFICATE

This is to certify that the students of Department of Statistics have successfully completed their research project entitled "Exploring Cancer Trends in India: A Statistical Approach to Gender and Zonal variation" with reference to requirements of "BSR".

This project was carried out as part of the academic requirements for the completion of their course and was conducted under the guidance of Dr. Moon, Head, Department of Statistics, Patna Women's College.

All three of the students, Priya Dhari Singh, Kesar and Sneha Kumari (2023-27) have demonstrated a comprehensive understanding of the research process and have employing appropriate statistical techniques and methodologies in accordance with the SRP guidelines.

We commend our students for their exceptional efforts and outstanding contributions to the field of statistical research.

SUPERVISOR
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ACKNOWLEDGMENT

With profound gratitude, we bow our head in reverence to **Almighty God,** whose constant presence, blessings and guidance have been the cornerstone of this journey. It is through his grace that we have been able to bring this research to fruition.

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Finally, to our family and friends, whose unwavering support, love and encouragement have been our anchor, we owe everything. This research stands as a collective effort, and we are deeply appreciative of each individual who has been part of it.

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ABSTRACT

Ever wondered how numbers quietly narrate the stories that shape a nation's health? Cancer statistics do exactly that—revealing patterns that often go unnoticed yet influence public health in profound ways. Behind every dataset lies a tale of resilience, disparity, and urgent need. This study steps into that silent world of data, decoding the evolving landscape of cancer in India through a statistical lens.

From Numbers to Narratives: Statistics often appear cold and distant, but in this context, they carry emotion and urgency. Each graph, rate, and comparison sheds light on unequal burdens—why certain zones report higher cases, why some cancers dominate among women while others affect men disproportionately, and how socio-economic and regional frameworks influence detection and outcomes. These insights push beyond simple description, offering direction for policy, awareness, and resource allocation.

From Patterns to Prevention: Cancer trends are more than counts on a chart—they are reflections of lifestyle shifts, environmental exposures, and genetic predispositions. By uncovering gender-based differences and zonal disparities, this study highlights the hidden dynamics influencing cancer incidence across the country. It combines rigorous data analysis with public health insight to reveal how demographic and geographical factors converge to shape cancer risk.

Cancer is not just a disease—it is a complex societal challenge influenced by culture, access, and awareness. Understanding its trajectory requires more than medical knowledge; it demands a statistical journey through India's diverse landscape. By exploring zonal

variation and gender disparities, this study reveals critical gaps and patterns that can guide future prevention strategies and healthcare planning.

This research examines India's cancer trends through comprehensive datasets, evaluating zonal variations, gender-based trends and age wise distribution of cancer cases in India. Through comparative analysis across regions and genders, it identifies emerging concerns, highlights disparities, and uncovers the underlying factors shaping the national cancer burden.

The findings aim to support policymakers, healthcare institutions, and public health practitioners with actionable insights grounded in statistical evidence. By transforming raw data into meaningful narratives, the study contributes to a deeper understanding of cancer's impact on India—helping illuminate the path toward early detection, equitable healthcare, and informed national strategy.

OBJECTIVE

This research aims to statistically analyze cancer incidence in India, focusing on the variations across gender, geographic zones and age distributions. The study seeks to identify key demographic, age, and zonal variations to understand how they shape the national cancer burden. The insights derived from this analysis are expected to support evidence-based planning, awareness strategies, and healthcare investments.

A comprehensive evaluation of gender-specific cancer patterns and zonal disparities will provide critical understanding of how lifestyle, socio-economic conditions, healthcare accessibility, and regional environments influence cancer incidence. By examining these interconnected variables through statistical tools, the research highlights the evolving landscape of cancer epidemiology in India.

This study is anticipated to offer a deeper perspective on the distribution and determinants of major cancer types across the country. By analyzing data from national reports, the project aims to uncover regional vulnerabilities—thereby emphasizing the importance of, early detection, and equitable healthcare distribution.

- To examine the spatial distribution of cancer incidence across various zones in India.
- To assess whether geographical zone of residence influences the probability of individuals being diagnosed cancer positive.
- To analyze whether gender plays a role in the number of cancers in India.
- To analyze the age group most at risk of developing cancer.

INTRODUCTION

India is a fast-growing economy experiencing rapid demographic and lifestyle changes. These shifts are starting to significantly affect the nation's disease patterns. Among the rising health issues, cancer has become a major public health threat, with increasing incidence in both urban and rural areas. As the economy develops, changes in diet, exposure to the environment, access to healthcare, and population structure are reshaping the cancer risk landscape in India. Understanding these trends, especially regarding gender, age and zonal differences, is crucial for developing prevention strategies, improving early detection, and informing healthcare policies. Cancer in India is now one of the biggest public health challenges of the 21st century. This issue is shaped by rapid urbanization, shifting lifestyles, environmental exposures, longer life expectancy, and a growing population. The burden of cancer has risen steadily over the past several decades, as reported by the National Cancer Registry Programme (NCRP) under the Indian Council of Medical Research (ICMR). Current estimates indicate that India sees over 1.4 million new cancer cases each year, with projections suggesting a continued increase. This rise is not just due to population growth; it also reflects improved diagnostic abilities, greater awareness, and changes in age demographics. India's age structure is shifting from a mostly young population to an older one. Since cancer risk increases with age, more cancers are being detected each year. The lifetime risk of developing cancer in India is about one in nine individuals by the age of seventy-five, highlighting the seriousness of the problem.

Basic Concepts and Nature of the Disease

Cancer constitutes one of the most complicated and variegated systems of diseases known to humanity, basically characterized by uncontrolled and abnormal growth of cells. It spares no one: it affects individuals across all ages, genders, nationalities, socio-economic levels, and educational backgrounds. The challenge about cancer is the variability in its presentation. It may present suddenly (acute onset), gradually over weeks or months (sub-acute), or even silently for many years (chronic). Even the symptoms vary tremendously from very nonspecific findings such as low-grade fever, chronic diarrhea, excessive fatigue, or unintended weight loss to more alarming findings like unusual bleeding, obstructive symptoms, palpable masses, or non-healing ulcers. Early recognition due to this variability often remains difficult, leading to delays in diagnosis.

Beyond its physical toll, cancer has massive psycho-social and economic burdens on both the patient and family. It affects their ability to work at an optimum capacity in personal and professional spheres, thereby placing heavy financial burdens on households and health systems. Societally, it is considered one of the major public health concerns for clinicians, health administrators, policymakers, and communities alike.

Causes and Risk Factors

The causes of cancer are multifaceted and frequently intertwined. While science has discovered numerous risk factors, most cancers still occur for which no specific cause can be identified. This uncertainty makes prevention efforts challenging but also underscores the value of

population-based data. Well-established contributors include genetic predispositions-meaning that some families carry inherited mutations that increase the risk of certain cancers. Environmental exposures are just as significant: the use of tobacco is among the highest risk factors for many cancers. Occupational exposures to damaging chemicals, pollution, radiation, and certain infections like HPV or hepatitis viruses are all big players.

Lifestyle factors comprise another broad category of risk. Poor diet, insufficient physical activity, excessive alcohol, high body mass index, diabetes, and metabolic disorders have been correlated with increased cancer susceptibility. In many instances, the risks depend on dose, duration, and temporal window of exposure-these diseases are more likely to appear after longer exposure, or exposure during particularly sensitive stages of life. These patterns come to light through epidemiological studies that enable governments and health systems to disseminate disease-specific preventive measures, like tobacco control, vaccination programs, and screening guidelines.

Zone based Cancer Distribution

Cancer distribution in India is highly varied, meaning some areas have much higher incidence rates than others. Western and Northern states often report elevated cancer rates, especially in certain districts where lifestyle habits, diets, and environmental exposures differ greatly from other regions of the country. Western India consistently reports some of the highest cancer rates for both men and women. Major cities like Delhi, Mumbai, Bengaluru, and Chennai also face significant cancer burdens, driven by pollution, sedentary lifestyles, later pregnancies

among women, and exposure to processed foods and occupational risks. Rural areas, historically under-diagnosed, are now beginning to show higher numbers as healthcare access improves.

Gender based Cancer Distribution

The cancer types most common in India differ between men and women. For men, cancers of the oral cavity, lungs, esophagus, stomach, and prostate are consistently among the most common. Oral and lung cancers in men are strongly linked to tobacco use, which remains widespread despite many public health campaigns. In India, tobacco is not just about smoking; smokeless forms like gutka, khaini, and paan with tobacco are common and significantly raise the risk of cancers of the mouth, tongue, throat, and esophagus. For women, the most prevalent cancers include breast cancer, cervical cancer, ovarian cancer, corpus uteri cancer, and lung cancer. Breast cancer has seen a particularly sharp increase in urban India, attributed to lifestyle changes such as delayed pregnancies, shorter breastfeeding periods, rising obesity rates, and physical inactivity. Although cervical cancer rates have declined over the years due to improved hygiene, screening, and awareness about HPV, it still remains one of the leading cancers among women, especially in rural and economically challenged areas.

Reasons for rise in Cancer Cases

The reasons for India's cancer burden are numerous and interconnected. Tobacco is the single largest preventable cause, accounting for about one-third of all cancer cases. Alcohol use, while

not as widespread, is increasingly linked to cancers of the liver, esophagus, and oral cavity. Unhealthy eating habits, including high salt intake, processed foods, red meat, and deep-fried foods, contribute to gastrointestinal cancers. Rapid urban growth has led to less physical activity, higher stress levels, rising obesity rates, and metabolic disorders like diabetes. All these factors are associated with cancers of the breast, colon, pancreas, and liver. Environmental aspects also play a significant role. Air pollution in large cities has been connected to lung cancer, even among non-smokers. Exposure to industrial chemicals, pesticides, contaminated water, and unsafe workplace conditions further raises risks, especially for vulnerable groups. Additionally, infections like human papillomavirus (HPV) and hepatitis B and C viruses continue to cause cervical and liver cancers, particularly in regions with limited vaccination and screening access.

Challenges Faced during Cancer Diagnosis

‘One major challenge India faces in addressing cancer is the delay in diagnosis for many patients. Many people seek medical attention only after symptoms become severe, at which point the cancer may be advanced. Factors contributing to late diagnosis include lack of awareness, fear of social stigma, limited availability of screening programs, low health literacy, and inadequate access to diagnostic facilities, especially in rural and tribal communities. Even when a diagnosis is made, accessing and affording cancer treatment can be difficult. While India has world-class oncology centers and skilled medical professionals, these resources are mostly found in urban areas, leading to disparities in care. Patients from rural areas often travel long

distances for treatment, creating financial and emotional burdens. The cost of cancer care—including surgery, radiation, chemotherapy, targeted therapies, and immunotherapies—can be overwhelming for families, leading many to face catastrophic health expenses.

Governments Initiative

Despite these challenges, India has made significant progress in creating policies and infrastructure to tackle cancer. The Government of India has launched the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular diseases and Stroke (NPCDCS). This initiative aims to promote early detection, strengthen diagnostic systems, expand treatment facilities, and integrate cancer care within primary healthcare. Large-scale screening programs for common cancers—breast, cervical, and oral—are being implemented. The introduction of the HPV vaccine has also created new possibilities for reducing cervical cancer rates, especially if vaccination coverage improves. Public campaigns focusing on tobacco control, promoting healthy lifestyles, and raising awareness about cancer symptoms are gaining ground, although their impact differs across socio-economic groups.

In summary, cancer in India is a growing and complex public health issue driven by demographic changes, lifestyle factors, environmental exposures, and systemic inequalities in healthcare access. While the country has made significant strides in cancer control through national programs, registry development, awareness campaigns, and expanded

treatment facilities, substantial challenges remain. Reducing the cancer burden in India will require a sustained, multi-faceted approach. This includes prevention through lifestyle changes, strict tobacco and alcohol controls, accessible screening and early detection services, affordable and decentralized treatment options, and ongoing, high-quality data collection. As India enhances its healthcare capabilities, today's efforts will be crucial in managing the rising cancer burden in the years to come.

NATIONAL CANCER REGISTRY PROGRAMME

An important initiative has been the strengthening of the National Cancer Registry Programme (NCRP). Since 1981, the NCRP has expanded significantly, now including numerous population-based and hospital-based cancer registries that collect data on cancer incidence, mortality, patterns, and survival. This information provides a basis for evidence-based policymaking. The data helps researchers identify trends, evaluate the effectiveness of public health interventions, and assess regional differences in cancer risk. It also allows India to contribute to international cancer databases and global comparative research. The growth of digital health systems, including electronic mortality records and hospital management systems, has further improved the quality of cancer data available.

Cancer Survival Trends

Cancer survival outcomes in India vary greatly depending on the type of cancer, stage at diagnosis, and access to treatment. Early-stage cancers have significantly better survival rates, yet many patients still present with advanced stages, lowering overall survival rates. This emphasizes the urgent need for widespread screening programs, public education, and improved referral systems. Childhood cancers, while relatively rare compared to adult cancers, have better treatment outcomes if diagnosed early. Leukemia is the most common childhood cancer in India, and many children can achieve long-term survival with prompt

intervention. However, there are still disparities in pediatric oncology services, with many children lacking access to specialized care.

Importance of Cancer Registries

Cancer registries form the basis for the understanding of the cancer pattern in any given population. A cancer registry is, in essence, an organized system for the collection, storage, analysis, and interpretation of data on individuals diagnosed with cancer within a defined area or healthcare institution. In the absence of such systematic collection of data, no realistic magnitude of cancer can be ascertained, nor its trends detected, nor effective control strategies planned. Cancer registries allow time-trend analysis, enabling researchers to determine whether the rates of cancers are increasing, decreasing, or staying stable. Registries also give an estimate of prevalence-the number of people living with cancer at any given point in time-a necessary count for healthcare resource planning.

However, collection of data at the population level is more efficient and cost-effective compared to large-scale intervention trials or long-term cohort studies. Thus, most countries worldwide depend predominantly on the data from cancer registries to plan their national programs for cancer control. Data from these registries provide evidence to design screening programs, resource allocation for treatment facilities, identify priorities, and plan public health interventions.

National Cancer Registry Programme in India

India was early to recognize the need for robust data on cancer and, therefore, set up the National Cancer Registry Programme in December 1981 under the Indian Council of Medical Research (ICMR). The coordinating center was set up in Bengaluru; over time, the programmed expanded into one of the largest cancer registry systems in the world. Today, the NCRP functions through the ICMR–National Centre for Disease Informatics and Research (NCDIR).

The NCRP encompasses information on cancer incidence, mortality, patterns, and geographic distribution in various regions of India. This information forms a basis for the development of policies on control and treatment in the setting of national cancer control programs, monitoring trends on cancer incidence, planning resources, and evaluating the effectiveness of ongoing programs. With the country still going through demographic changes-increased life expectancy-there has been a continuous rise in the number of cases, as depicted by registry data. This increase is partly because the elderly population is on the rise, among whom the risk of cancer is naturally very high.

Objectives and Functions

The NCRP had several broad and interlinked objectives when it came into being. First, it looks to generate data on the magnitude and patterns of cancer across India, which includes delineation by region, gender, and age group. Second, these data form a basis for further epidemiological research to help scientists identify emerging risk factors and new trends. Third, the NCRP supports the planning and evaluation of cancer control activities under national programs such as the National Programme for Prevention and Control of Cancer, Diabetes,

Cardiovascular Diseases and Stroke. Last but not least, it helps build capacity through the provision of training in cancer registration and epidemiology to health workers and researchers throughout the country.

From the six registries that it began with-including three population-based cancer registries in Bangalore, Chennai, and Mumbai and three hospital-based registries in Chandigarh, Dibrugarh, and Thiruvananthapuram-it has grown manifold to include 36 PBCRs and 236 HBCRs to date, thus reflecting India's heightened response to cancer surveillance.

Types of Cancer Registries in India

NCRP maintains two major kinds of cancer registries:

a. Population-Based Cancer Registries - PBCRs

PBCRs collect data on all new cancer cases occurring within a clearly defined geographic population. Because they aim to capture every case in the specified area—regardless of where the patient is treated—they provide the most accurate picture of cancer incidence and patterns in the community. The PBCRs will be important for understanding regional variations and the identification of high-risk groups. They also generate data that can be compared with global registries to evaluate India's standing in international cancer statistics.

b. Hospital-Based Cancer Registries (HBCRs)

HBCRs have their main focus on patients treated at a given hospital. Diagnosis, treatment, and outcomes data are recorded prospectively,

thus enabling a useful insight into clinical management. Although they cannot supply general population incidence information, the role of HBCRs in the assessment of the effectiveness of treatment, survival, and workload of the hospital is evident. They are also used as source material by clinicians who may want to study any trends that may emerge in the profile of patients or response to certain treatments.

Apart from these, Cancer Atlas methods are used for short-term studies or mapping of cancer hotspots across different regions, especially in areas with limited registry coverage.

Data Collection Challenges and Methodology

A significant challenge in India is that cancer is not a logistically notifiable disease. This implies that registries cannot depend on spontaneous reporting by hospitals and healthcare centers. Thus, staff at the registry do active data collection by visiting hospitals, clinics, pathology laboratories, imaging centers, and even local government offices responsible for issuing death certificates. They manually extract relevant information about cancer cases and deaths where cancer is mentioned as the cause. The whole process is intensive, needing skilled manpower, huge coordination, and stringent quality-control measures.

The data collected enter into specialized software prepared by ICMR–NCDIR. Using leading-edge information technologies, databases check for accuracy, duplicate elimination, consistency checks, and record linkage of mortality with incidence records. Standardized training programs are provided on a regular basis to maintain quality standards. These registries interact with healthcare providers at the grassroots

level, administrative departments, and communities to establish collaborative relationships that make the process of collecting the information easier.

Importance and Use of NCRP Data in Public Health and Research

The data generated by the NCRP are of immense value for the planning of health services. It could help in decisions like where to open new cancer treatment facilities, in which populations awareness needs to be increased, and where the financial investments for chemotherapy or radiation therapy facilities have to be made. For clinicians, registry data offer insights into prevailing cancer trends, treatment outcomes, and survival patterns across India. Researchers depend on these data to conduct epidemiological studies, assessments of treatment efficacy, and identification of emerging cancers or changes in the disease burden.

Importantly, these contribute to international publications like CI5 (Cancer Incidence in Five Continents) by IARC, allowing global comparison. These collaborations have enriched India's budding capacity for cancer surveillance and will add to the evidence-based decision-making globally.

Population Estimation for Cancer Calculations

Accurate population data are essential for the calculations of cancer rates. The census data of 2001 and 2011 in India are used for projection to estimates for subsequent years by the difference distribution

method. This ensures that the incidence and mortality rates are derived from realistic populations, which is particularly important in countries experiencing significant demographic changes.

Accordingly, the emerging burden of cancer in India reminds us of an urgent need for a coordinated and sustained national response that accords equal priority to prevention, early detection, treatment, and long-term patient support. The diversity of cancer patterns is evident across regions, between genders, and within various socioeconomic groups, reflecting a complex interplay of lifestyle factors, environmental exposures, genetic predispositions, and inequalities in access to health care. This rising incidence partially reflects demographic transitions and improved diagnostic capabilities, but also partly signals the presence of growing gaps in awareness, screening coverage, and access to timely medical care. The significant progress represented by expanding the National Cancer Registry Programme, developing national cancer control strategies, and investing in diagnostic and treatment infrastructure should be supplemented with strong public health education, more stringent tobacco and alcohol policies, wider vaccination and screening efforts, and enhanced affordability and decentralization of oncology services. Cancer in India needs to be tackled not only with medical and technological interventions but also with social, behavioral, and policy changes that will alter the way communities across the country deal with the disease. Only through a comprehensive, data-driven, and equitable approach will it be possible for India to reverse the current trend, reduce preventable cancers, improve survival outcomes, and eventually mitigate the profound impact of cancer on individuals, families, and the nation as a whole.

METHODOLOGY

The study on Exploring Cancer Trends in India: A Statistical Approach to Gender and Zonal variation was conducted under the academic guidance of Patna Women's College, an Autonomous Institution under Patna University, NAAC Reaccredited with an A++ Grade in 4th cycle, and recognized as a "College with Potential of Excellence (CPE)" by the UGC.

This study examines the incidence of cancer cases between gender distribution (males and population) across different age group based on the zonal variations. The analysis employs both the chi square test of Independence and Simple Linear Regression to understand association and predictability. The following methodological steps were undertaken:

1. Defining the population:
    - The population for this research consists of dataset provided at the National Centre for Disease Informatics and Research.
    - The entire dataset is divided into 6 zones namely North East Zone, North Zone, South Zone, East Zone, West Zone and Central Zone.
    - Under each of these Zone there are multiple states which are further categorized by the gender corresponding to the age-group they belong to.
    - Using the data from the NCDIR website, we obtained officially verified figures for males and females population across zones, ensuring that the study is based on credible and authenticated dataset required for the research work.

2. Data Collection

The present research was conducted entirely using secondary data, ensuring reliability, accuracy and consistency of information. The dataset used in the research was sourced from the National Centre for Disease Informatics and Research (NCDIR), an authoritative government body responsible for maintain comprehensive and standardized disease-related records in India. The data obtained from this official website is systematically collected, rigorously validated, and widely used for epidemiological research and public health planning.

Relying on secondary data from NCDIR provided several advantages for the study. First, it ensured access to large-scale population level information covering multiple regions, age-groups and gender categories, which would not be feasible through primary data collection. Second, the use of government-published data enhanced the credibility and authenticity of the analysis, as the figures are based on structured reporting mechanisms and standardized cancer registries. Third, the data offered a consistent framework for comparison across zones enabling meaningful statistical analysis and interpretation.

By using secondary data from a nationally recognized institution, the study minimized potential biases associated with self-reported or sample-based data and ensured that the findings reflect actual reported cancer incidence patterns. This approach makes the results actual reported cancer incidence patterns. This approach

makes the results suitable for policy-oriented discussions and supports evidence-based decision making in cancer control, healthcare planning and resource allocation.

3. Understanding the data

The dataset used for this study is comprehensive in nature and is structured to capture cancer incidence across multiple geographic and demographic dimensions. It encompasses six major zones, namely the North East Zone , North Zone, South Zone, East Zone, West Zone and Central Zone, each representing distinct regional groupings within the country. This zonal classification enables a clear comparison of cancer patterns across different geographic areas and facilitates the identification of regional disparities.

Within these zones, the data further disaggregates information at the state level. The North East Zone includes five states- Manipur, Mizoram, Sikkim, Meghalaya and Tripura- which together represent a region known for its unique demographic, environmental, and healthcare characteristics. The North Zone consists of Punjab and Delhi with Delhi additionally serving as a major healthcare hub that attracts patients from neighboring regions. The South Zone includes three states – Telangana, Tamil Nadu and Karnataka all of which have relatively well-established health care infrastructure and cancer treatment facilities. The East Zone is represented by West Bengal comprising Gujarat and Maharashtra, both of which are significant contributors to the overall cancer burden due to their large population and advanced diagnostic capabilities. Finally, the Central Zone includes Madhya

Pradesh, representing a large geographic area with diverse urban and rural healthcare access.

Such a detailed and hierarchical dataset supports the application of various statistical techniques, including descriptive analysis, chi square testing, correlation and regression analysis. Overall, the structure of the data enabled a comprehensive understanding of cancer distribution patterns and provides a strong foundation for meaningful interpretation and evidence-based conclusion.

4. Statistical Methodology

   i. Correlation analysis:

      Statistically, correlation analysis can be described as a method through which the strength and direction of a relationship between variables are to be measured. It helps in determining patterns and associations, which then allows researchers to understand the relationship between changes in one variable and changes in another. Correlation analysis is used to determine the degree and direction for the relationship of two or more variables. Its most commonly used in the following way:

      First is the necessary data for the variables are obtained. We collect two sets of data that we are interested in analyzing. For instance, let's say we have two variables:

      - X
      - Y

Each variable has multiple data points. Let (X_1, Y_1), (X_2, Y_2), ,……., (X_n, Y_n) be pairs of values.

Formula for Pearsons Correlation Coefficient:

$$\text{Corr (r)} = \frac{\sum(x-\bar{x})*(y-\bar{y})}{\sqrt{(\sum(x-\bar{x})\text{^}2)*(\sum(y-\bar{y})\text{^}2)}}$$

where,

n= Number of data points

X and Y are the two variables being compared

Pearsons's correlation coefficient ranges from -1 to +1 but can take any value in between. A positive correlation means that the graph slopes up, as one variable increase so does the other. A negative correlation means that the graph slopes down, as one variable increases the other decreases.

ii. Simple Linear Regression:

Simple Linear Regression is a method of applying statistical techniques on the relation between one independent variable and one dependent variable, assuming a linear relationship and that every change in independent variables corresponds to a similar change in dependent variable. Simple Linear Regression is used to yield estimates of coefficient, which imply the strength and direction of those relationships. SLR finds applications for predicting values and understanding trends and analyzing variables effects on an outcome.

- Dependent variable (Y): The phenomenon we want to predict
- Independent variables (X): Predictor or factors that affect the dependent variable

General formula for SLR is:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

where,

Y is the dependent variable

β_0 is the intercept

β_1 is the coefficient of the independent variables X

ε is the error term

The coefficients are determined through a process called least squares estimation, which minimizes the sum of squares differences between the observed and predicted values.

Once we have the coefficients, we calculate the predicted values $\hat{Y}$ using the formula:

$$\hat{Y} = \beta_0 + \beta_1 X$$

Residuals are the differences between the observed values and the predicted values:

$$\text{Residuals} = Y - \hat{Y}$$

After determining the coefficient and assessing the model, we use the regression equation to make predictions on new data by plugging in values for the independent variables.

iii. Hypothesis Testing and $\mathbf{X^2}$ statistics:

A hypothesis is a statement or an assumption of a relationship between two or more variables in research. It acts as a foundation for testing and analysis, giving a direction to the study. Hypotheses are made to predict an outcome, explaining phenomena, or test existing theories.

Null Hypothesis:

The null hypothesis is a statement of no effect between variables. It states that the observed effect within the data is due to random chance or natural variation. Researchers test this to see whether it can be rejected or not.

The null hypothesis states there is no significant relationship or effect between the variables studied.

Alternate Hypothesis:

The alternate hypothesis is a statement that denies the null hypothesis. It means that there is a significant effect between the variables that are being studied. The alternative hypothesis is the assumption that there is a meaningful and measurable effect or relationship between the variables being investigated.

For our research we set 3 hypotheses:

1. **To analyze whether gender plays a role in the number of cancers cases in India.**

   - H_0 (Null Hypothesis)
     There is no difference between the cancer cases of males and females
   - H_1 (Alternative Hypothesis):
     There is a difference between the cancer cases of males and females

2. **To assess whether geographical zone of residence influences the probability of individuals being diagnosed cancer positive.**
   - H_0 (Null Hypothesis)
     Cancer cases are equally distributed across all zones
   - H_1 (Alternative Hypothesis):
     Cancer cases are not equally distributed across all zones
3. **To assess whether geographical zone of residence influences the probability of individuals being diagnosed cancer positive.**
   - H_0 (Null Hypothesis)
     Cancer cases are equally distributed across all zones
   - H_1 (Alternative Hypothesis):
     Cancer cases are not equally distributed across all zones

The chi square test is a non-parametric statistical method used to test the hypothesis of independence between categorical variables by measuring the discrepancy between observed and expected frequencies under the assumption of no association. It is given by the formula:

$$X^2 = \sum \frac{(\boldsymbol{Observed\ Frequency - Expected\ Frequency})\text{^}\mathbf{2}}{\boldsymbol{Expected\ Frequency}}$$

DATA ANALYSIS

1. Which zone has the highest incidence of cancer cases in India?

In this section we will dive deep into the zone that has highest number of cancer cases in India. In the dataset, India is divided into 6 zones namely Central Zone, East Zone, West Zone, North East Zone, North Zone and South Zone. The below analysis is providing a detailed knowledge of which zone has the highest incidence of Cancer cases in India based on the provided dataset.

Tabular Depiction:

Zones	Total
Central Zone	41254260
East Zone	89541595
North East Zone	54540323
North Zone	157220198
South Zone	103419407
West Zone	223443845

Table 1.1: Zonal distribution of Cancer Cases

Graphical Visualization:

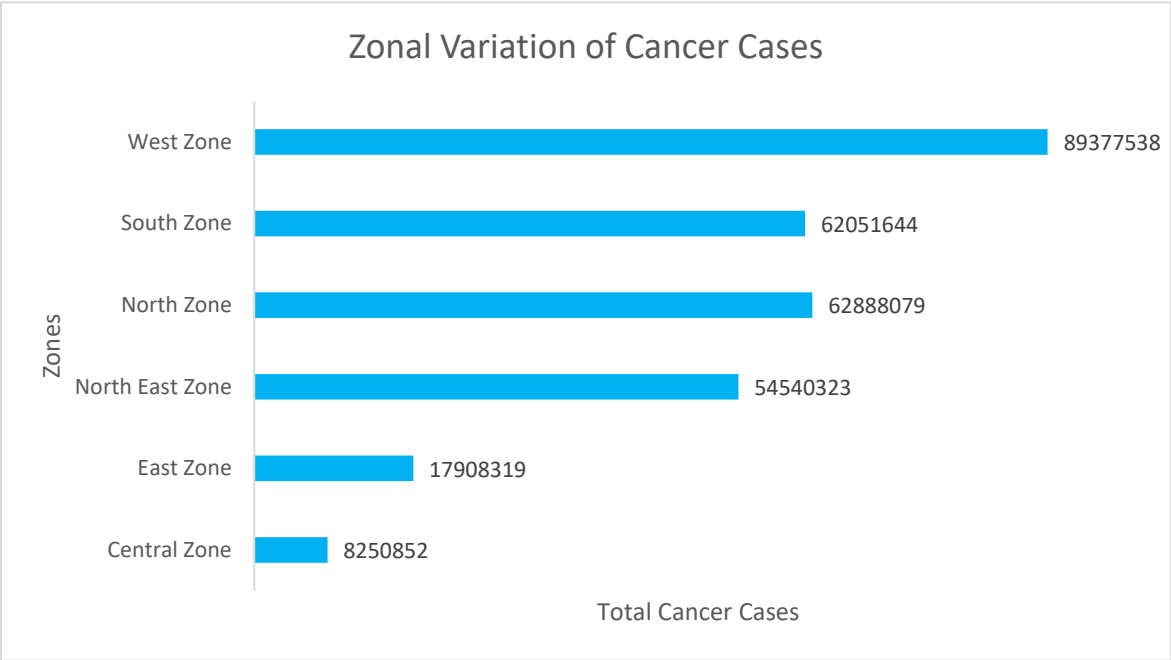

Fig 1.1: Graphical Representation of Zonal distribution of Cancer Cases

Analysis and Interpretation:

Fig 1.1 illustrates a clear variation in the total number of cancer cases across different zones, highlighting substantial geographic disparities. The West Zone stands out prominently as the region with the highest cancer burden, reporting nearly 89 million cases far exceeding all other zones. Following this, the North Zone and South Zone show similarly elevated case counts, each exceeding 62 million, indicating that these regions also contribute significantly to the overall cancer load. The North East Zone occupies a middle position with approximately 54 million cases, reflecting a moderate level of incidence relative to the higher burden zones. In contrast the East Zone reports just 18 million cases showing considerably fewer cases than the major high burden regions. The Central Zone with around 8 million cases,

records the lowest number of cancer cases among all zones. Overall, the distribution is highly uneven, with certain zones particularly the West, North and South experiencing disproportionately higher cancer prevalence.

2. Which state in each zone has the highest cancer cases?

In fig 1.1 we saw the zones and the number of cancer cases in each zone. Now every zone comprises of multiple states. In this section we will find which state has the highest number of cancer cases.

i. North East Zone:

The north east zone consists of 5 states enumerated for total cancer cases registration (males + females) namely: Tripura, Meghalaya, Sikkim, Mizoram and Manipur.

Tabular Depiction:

Region	Total
MANIPUR	15778153
MIZORAM	5862530
SIKKIM	3324050
MEGHALAYA	10019395
TRIPURA	19556195

Table 1.2: Cancer cases in Regions of North East Zone

Graphical Visualization:

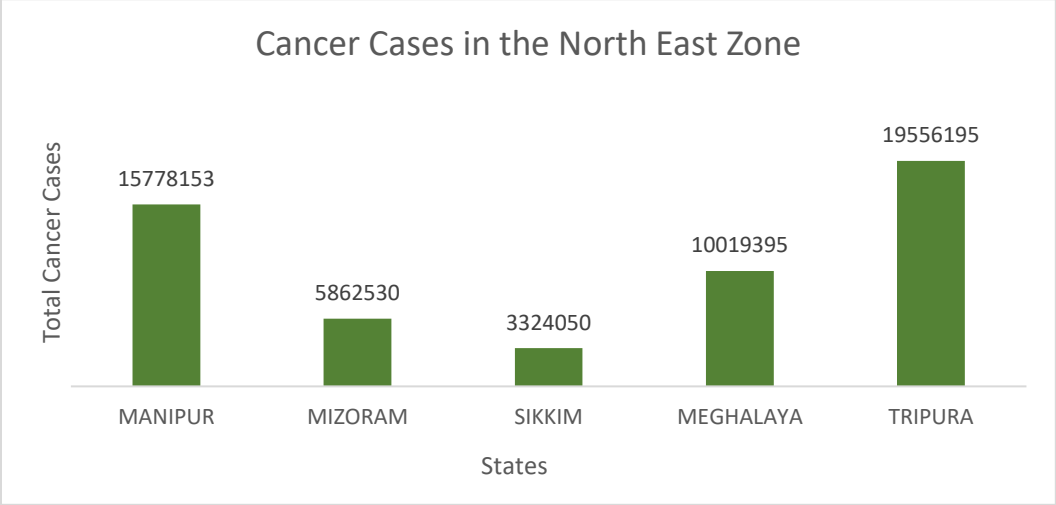

Fig 1.2: Graphical Representation of Cancer Cases in Regions of North East Zone

Analysis and Interpretation:

The data highlights the total number of cancer cases registered across five North Eastern states, revealing clear differences in their demographic sizes. Tripura has reported the highest number of cases, indicating a significantly greater cancer burden compared to the other states. Manipur follows with a substantial number of registered cases, through still notably lower than Tripura, suggesting a high but relatively lesser disease load. Meghalaya shows a moderate level of cancer incidence, falling in the mid-range among all states.

Mizoram and Sikkim record comparatively lower numbers of cases, implying a smaller registered cancer burden within

their populations. These differences suggest that the impact of cancer varies widely across the region, influenced by demographic, environmental and healthcare accessibility factors unique to each state. Such findings highlight the need for state- specific cancer control strategies, ensuring that resource allocation, screening programs and treatment support are aligned with actual cancer incidence in each area. This targeted approach would strengthen cancer management and improve overall public health outcomes across the region.

ii. West zone:

The west zone consists of 2 states for total number of cancer cases (males + females) namely: Mumbai and Gujarat.

Tabular Depiction:

Region	Total
GUJARAT	28168662
MAHARASHTRA	122417752

Table 1.3: Cancer cases in Regions of West Zone

Graphical Visualization:

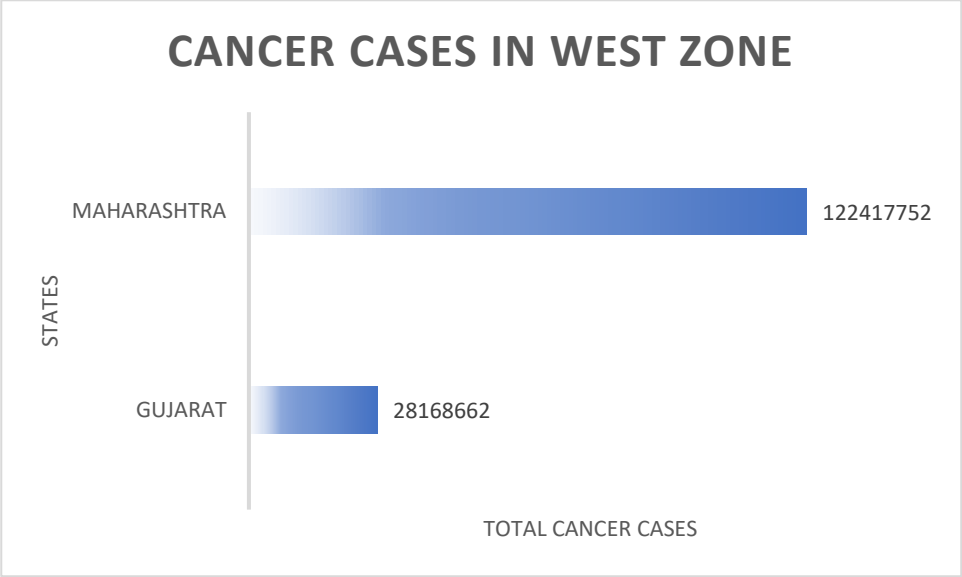

Fig 1.3: Graphical Representation of Cancer Cases in Regions of West Zone

Analysis and Interpretation:

The data presents noticeable variations in the number of cancer cases recorded across different centers. Maharashtra reports the highest number of registered cases, indicating a significantly larger cancer burden and suggesting a high patient inflow or extensive reporting coverage within the region. Gujarat follows with a substantial number of cases, reflecting a major load on its healthcare services, through still considerably lower than Maharashtra. Such high values of cancer cases in Maharashtra can also be due to people moving to Mumbai in order to get efficient cancer treatment.

These differences highlight the diverse cancer incidence patterns across the centers. Such variation underscores the

need for center -specific strategies, with stronger support, improved diagnostics and tailored interventions in higher-burden areas like Mumbai, Ahmedabad, and Pune, while ensuring adequate outreach and awareness efforts in lower-reporting regions.

iii. Central Zone:

The central zone comprises of 1 state namely Madhya Pradesh. The following graph and table is an indication of number of cancer cases in this zone.

Tabular Depiction:

Region	Total
MADHYA PRADESH	8250852

Table 1.4: Cancer cases in Regions of Central Zone

Graphical Visualization:

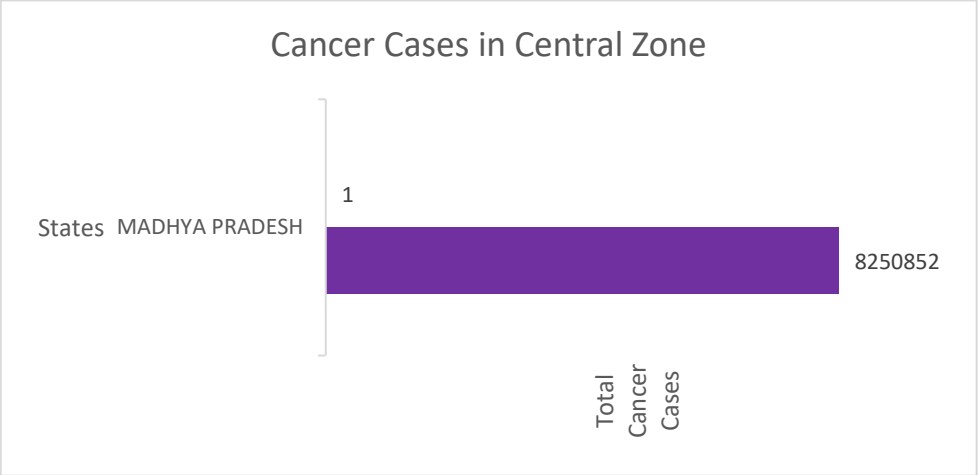

Fig 1.4: Graphical Representation of Cancer Cases in Regions of Central Zone

Analysis and Interpretation:

The data indicates the Madhya Pradesh has reported a substantial number of cancer cases with over 8.2 million registered cases, reflecting a significant disease burden within the state. Although this value is lower compared to some higher burden regions, it still represents a considerable patient volume that places pressure on the staes healthcare system. Several factors may contribure to htsi reproted figure, including the states large population base, limited awareness of early symptoms, delayed diagnosis, and varying levels of access to specialized cancer care services across urban and reral areas.

The magnitude of cases underscores the importance of strengthening healthcare infrastructure and enhancing regional caner traetment capactiry within madhaya Pradesh.

This finging highlighrs the need for imptoved public health stratergies such as early detection programs, widespread awarness initiatices and more accessible screening services, especially in underserved districts. Continued investment, focused government intiatives, and coordinated institutional efforts will be essential to effectively address the cancer burden and ensure better health outcomes for communities across the state.

iv. East Zone:

The east zone comprises of 1 state namely West Bengal. The following graph and table are an indication of number of cancer cases in this zone.

Tabular Depiction:

Region	Total
West Bengal	17908319

Table 1.5: Cancer cases in Regions of East Zone

Graphical Visualization:

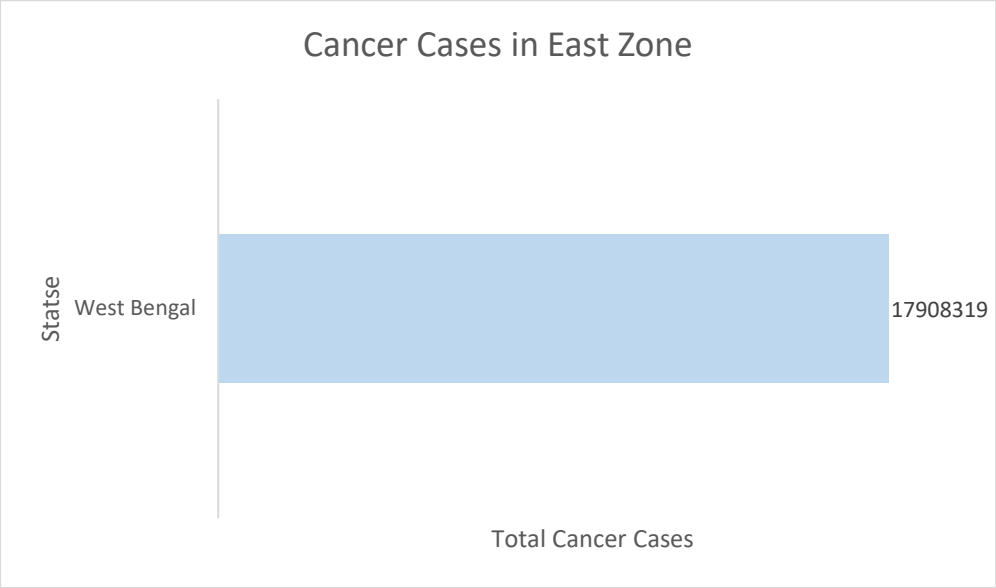

Fig 1.5: Graphical Representation of Cancer Cases in Regions of East Zone

Analysis and Interpretation:

The data indicated that West Bengal has recorded a very high number of cancer cases, reflecting a substantial disease burden within the region. With over 17 million registered cases, West Bengal stands out as a major center with significantly patient volume. This large figure may be influenced by several factors, including a dense population, advanced diagnostic facilities and greater accessibility to cancer care services, all of which contribute to higher case reporting.

The magnitude of cases highlights the importance of strong healthcare infrastructure, well-equipped cancer treatment centers, and comprehensive public health strategies in West Bengal. The finding emphasizes the

need for continued investment in early detection, awareness campaigns, and screening programs to manage the high cancer load effectively. Tailored interventions and sustained government and institutional support will be crucial to address the growing challenges and ensure better health outcomes for the population.

v. South Zone:

The south zone consists of 3 states namely Telangana, Tamil Nadu and Karnataka. The following graph and table are an indication of number of cancer cases in this zone.

Tabular Depiction:

Region	Total
TELANGANA	11981206
TAMIL NADU	23762760
KARNATAKA	26307678

Table 1.6: Cancer cases in Regions of South Zone

Graphical Visualization:

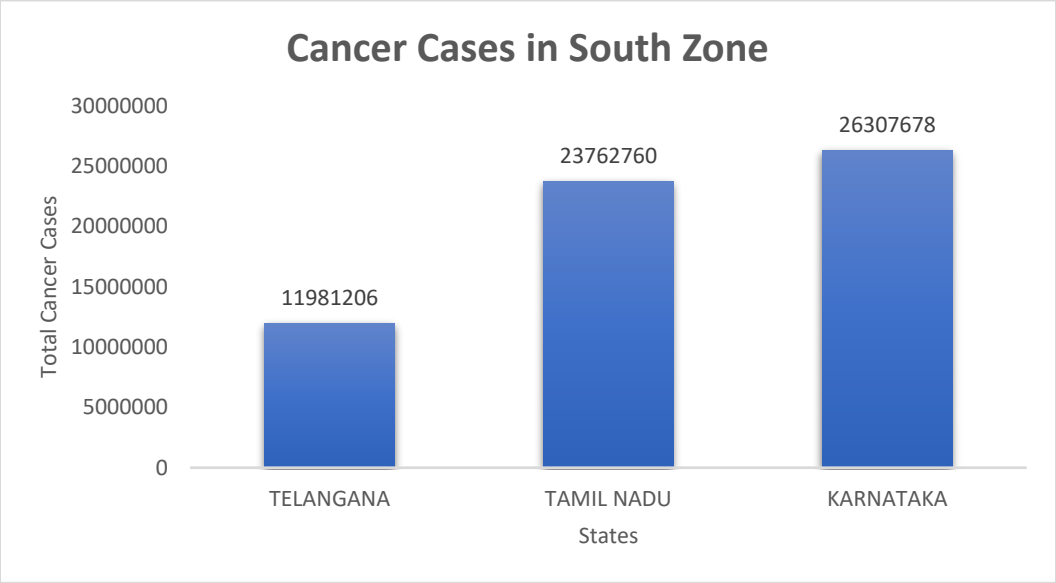

Fig 1.6: Graphical Representation of Cancer Cases in Regions of South Zone

Analysis and Interpretation:

The data highlights notable differences in the number of cancer cases reported across the three metropolitan centers. Karnataka records the highest total number of cases, indicating a substantial cancer burden and reflecting its large population base, advanced healthcare infrastructure, and wider diagnostic coverage. Tamil nadu follows closely, with a comparably high number of registered cases, suggesting similar levels of reporting and healthcare accessibility.

Telangana shows a significantly lower number of registered cases in comparison to both Karnataka and Tamil Nadu. This

may be due to differences in population size, reporting practices, availability of cancer registries or diagnostic reach within the region.

These variations emphasize the differing levels of cancer incidence and detection across major Indian cities. Centers with higher cases load such as Karnataka and Tamil Nadu, may require strengthened capacity in treatment facilities early detection programs, and public health initiatives. Meanwhile, Telangana may benefit from enhanced awareness, expanded screening efforts, and improved reporting systems to ensure comprehensive cancer surveillance.

vi. North Zone:

The north zone consists of 2 states namely Punjab and Delhi. The following graph and table are an indication of number of cancer cases in this zone.

Tabular Depiction:

Region	Total
PUNJAB	10065059
DELHI	51923020

Table 1.7: Cancer cases in Regions of North Zone

Graphical Visualization:

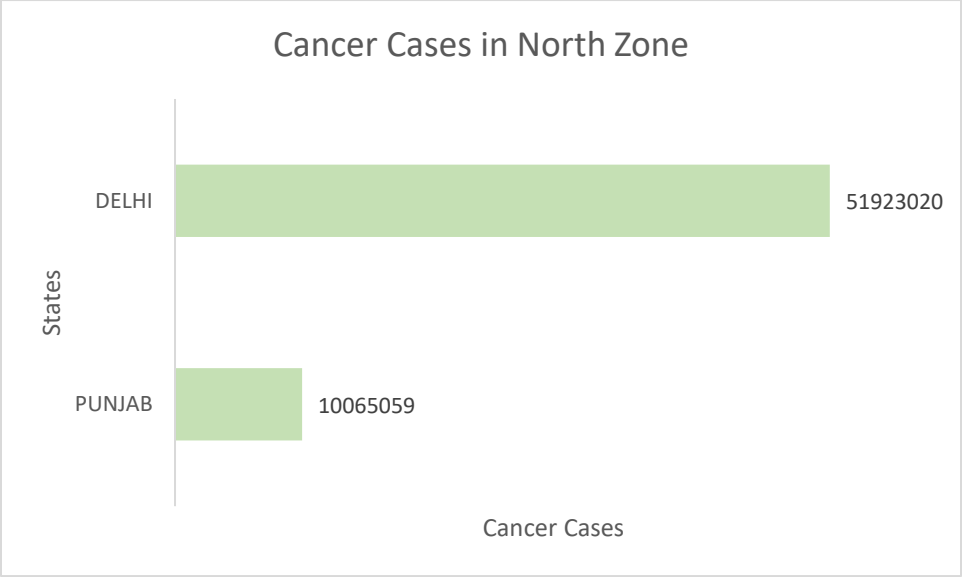

Fig 1.7: Graphical Representation of Cancer Cases in Regions of North Zone

Analysis and Interpretation:

The data indicates that Delhi has recorded a significantly higher number of cancer cases compared to Punjab, reflecting a notably larger disease burden within the region. With over 51 million registered cases, Delhi stands out as a major center with exceptionally high patient volume. Such a large figure may be influenced by several contributing factors, including a dense and highly urbanized population, the presence of advanced diagnostic facilities, and greater accessibility to specialized cancer care. Delhi also serves as a major healthcare hub, attracting patients from neighboring states, which can further increase the total number of reported cases.

In contrast, Punjab has recorded around 10 million cases, which, although substantial, remains far lower than Delhi's total. This difference may be shaped not only by population variations but also by differences in health-seeking behavior, availability of screening programs, and the extent of reporting and documentation systems. Lower reporting capacity or fewer specialized centers may result in fewer cases being captured in official statistics.

The magnitude of Delhi's cancer burden highlights the need for strengthened healthcare infrastructure, efficient referral systems, and well-coordinated public health strategies to manage the high patient load effectively. Investments in early detection, community-based awareness programs, and preventive health initiatives will be crucial for improving outcomes. Similarly, Punjab can benefit from expanded screening networks and improved access to diagnostic services to ensure timely identification and treatment. Both regions will require sustained government support and data-driven interventions to address the rising challenges and protect population health.

3. Which age group is highly prone to cancer

This analysis explores how cancer cases are distributed across age groups in India, highlighting the age ranges where incidence is highest. Identifying these patterns help guide prevention, early detection and resource planning.

Tabular Depiction:

Age Group	Total
00-04	23456241
05-09	24373027
10-14	25635098.5
15-19	26876491
20-24	30442587
25-29	29320648.5
30-34	25468697
35-39	23097636
40-44	19860908
45-49	17065502.5
50-54	13595898
55-59	10586263.5
60-64	8912146
65-69	5981238
70-74	4558818.5
75+	5785554.5

Table 1.8: Age wise distribution of Cancer Cases

Graphical Visualization:

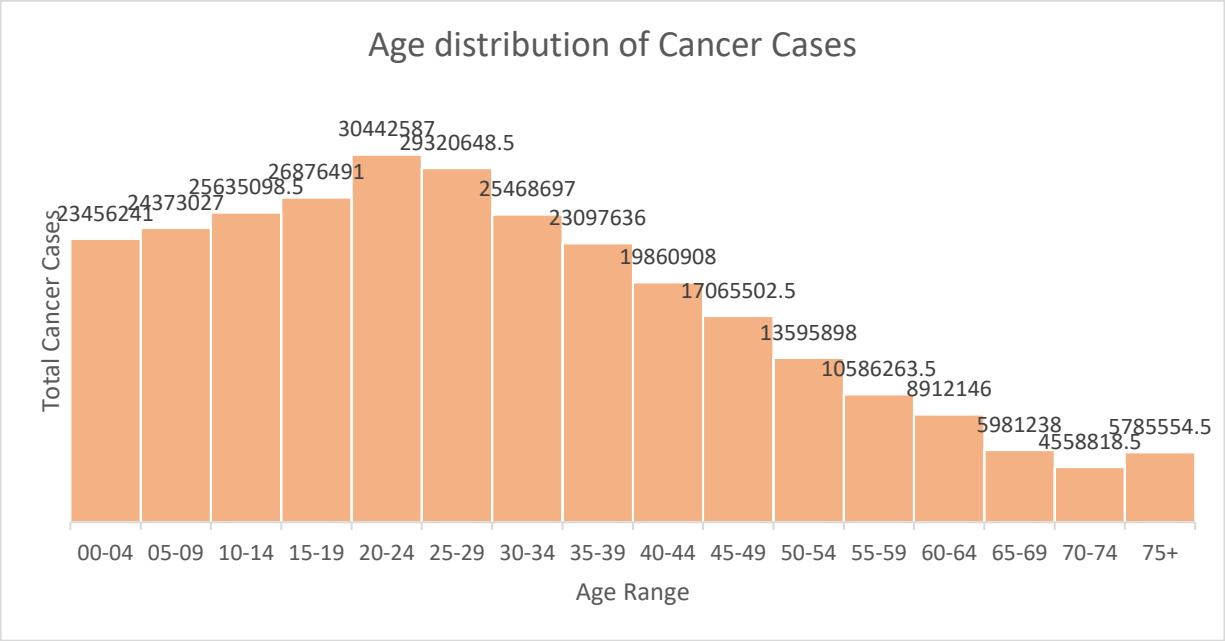

Fig 1.8: Graphical Representation of Age wise distribution of Cancer Cases

Analysis and Interpretation:

The data reveals a distinct pattern in cancer incidence across different age groups. There is a steady increase in the number of cases from childhood toward early adulthood, with cases rising consistently from the 00-04 group up to the 20-24 age group. The highest number of cases is observed in the 20-24 age group, indicating a peak in early adulthood.

Following this peak, the cases remain high in the 25-29 and 30-34 age groups, after which a gradual decline begins. The reduction becomes more pronounced from the 40-44 age group onward, with cases steadily decreasing through the middle-aged and older populations. The lowest incidence appears in the oldest age groups 70-74 and 75+ suggesting a significant downward trend in later years.

This pattern highlights that the burden of cancer is disproportionately higher among younger and middle-aged adults, particularly those in their early twenties. These findings underscore the need for targeted awareness, screening initiatives, and early-age interventions to address the higher risk observed in younger populations.

4. Which gender is most at risk of developing cancer

The analysis explores which gender shows a higher likelihood of being diagnosed with cancer. By assessing male and female

case trends, it highlights disparities in risk and helps identify groups requiring focused prevention and screening measures.

Tabular Depiction:

Gender	Total
Males	152100186
Females	142916569

Table 1.9: Total cancer cases against the gender

Graphical Visualization:

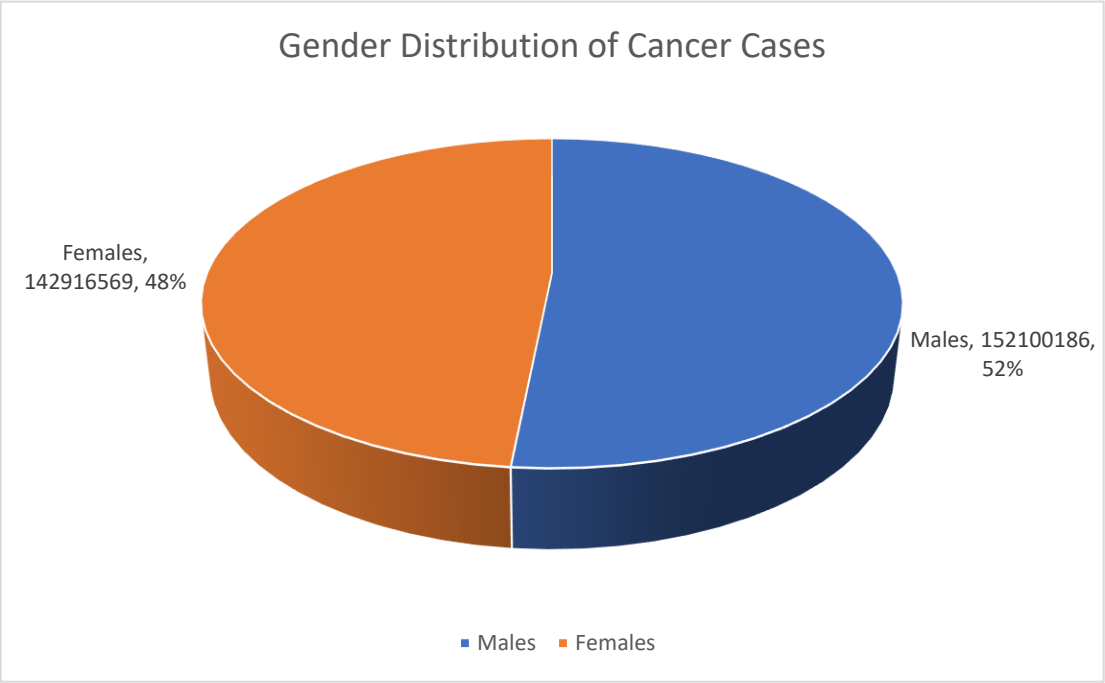

Fig 1.9: Graphical Representation of Cancer cases against gender

Analysis and Interpretation:

The nearly equal gender distribution emphasizes that cancer is a shared health burden across the population — not concentrated heavily in one gender. This underlines the importance of ensuring equitable access to cancer screening, diagnostic, and treatment services for both men and women. Furthermore, because neither group is dramatically overrepresented, public health strategies should target both genders with awareness, prevention, and care interventions.

The relatively balanced ratio also signals that other factors — such as age, geographic zone, environmental exposure, lifestyle, or socioeconomic status — might play a larger role in influencing cancer incidence than gender alone. As a result, a comprehensive approach that addresses multiple risk factors, while ensuring gender-neutral healthcare access, is crucial for effectively managing the cancer burden in the population.

5. Region wise gender analysis of cancer patients

This section provides an information about the number of cancer cases distributed with respect to the gender (males and females) in each zone. It is very important to understand the effect of cancer on both males and females population in a particular zone. Understanding these patterns will help drive correct government policies and better healthcare infrastructure.

Tabular Depiction:

Zones	Males	Females
Central Zone	4280917	3969935
East Zone	9270946	8637373
North East Zone	27541445	26998878
North Zone	32929570	29958509
South Zone	31641265	30410379
West Zone	46436043	42941495

Table 2.0: Gender wise distribution of Cancer cases in every zone

Graphical Visualization:

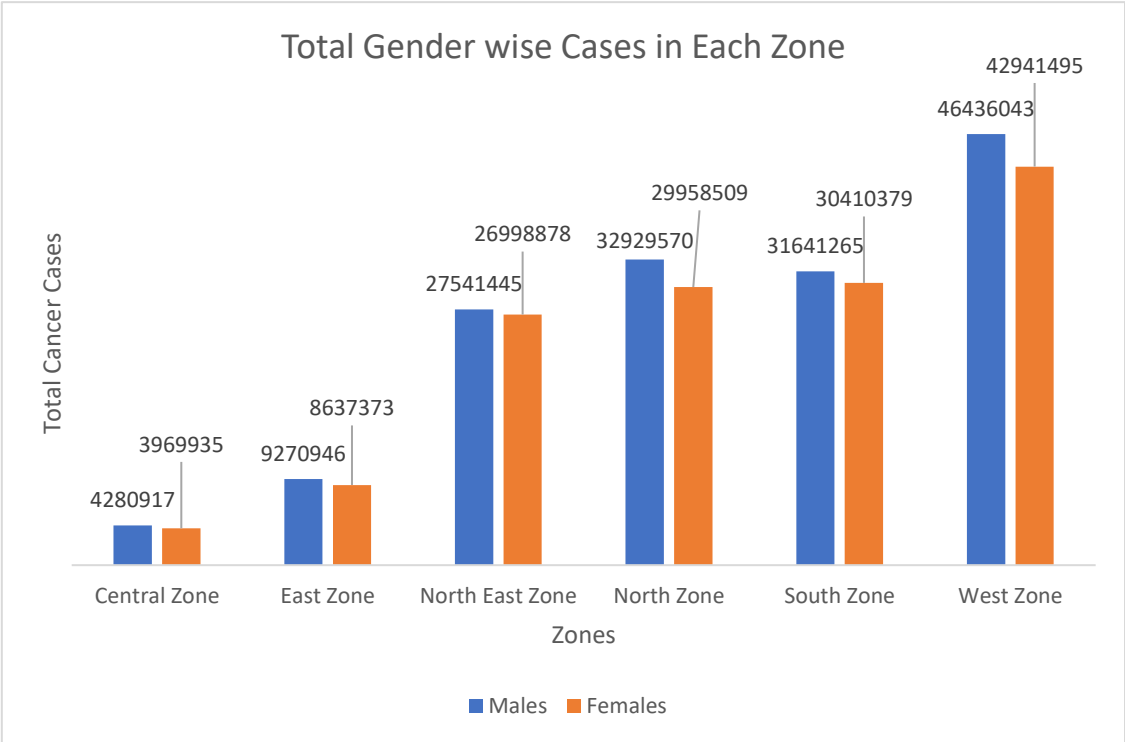

Fig 2.0: Graphical Representation of Gender wise distribution of Cancer cases in every zone

Analysis and Interpretation:

The fig 2.0 presents a comparative view of total cancer cases among males and females across different zones, revealing distinct regional patterns as well as gender differences in reported cases. Overall, male cases are consistently higher than female cases in every zone, although the gap varies in magnitude from one region to another.

In the Central Zone, both male and female counts are relatively low, with males at around 4.28 million and females at about 3.96 million. This suggests a comparatively smaller cancer burden in this zone. Moving to the East Zone, the numbers increase but still remain moderate, with males recording 8.6 million cases and females around 9.2 million. Interestingly, this is the only zone where female cases slightly exceed male cases, which may indicate differences in reporting, screening participation, or regional lifestyle patterns.

A substantial rise appears in the North-East Zone, where both genders show more than 26–27 million cases. The similar magnitude for both groups reflects a more evenly distributed burden, possibly indicating consistent detection processes or comparable exposure levels across genders.

In the North Zone, totals climb even higher, crossing nearly 29–30 million cases for both males and females. The region seems to experience a significant cancer load, and the difference between

genders narrows even further, indicating near-parallel trends in disease occurrence.

The South Zone shows a similar pattern, with both groups around 30–31 million cases. This steady rise highlights a substantial healthcare demand across the southern states, necessitating robust screening, diagnostic, and care capacity.

The West Zone stands out with the highest reported cancer cases, recording more than 46 million males and 42 million females. This large volume can be attributed to higher population density, better reporting systems, or broader access to diagnostic facilities, leading to more accurate case registration.

Across all zones, the consistent observation is that cancer affects both males and females at large scale, with males maintaining a slight lead in total cases. The data underscores the importance of zone-specific cancer surveillance, resource allocation, and targeted public health strategies. Regions with exceptionally high numbers—such as the North, South, and West—may require intensified screening campaigns, expanded treatment infrastructure, and tailored awareness programs to address their substantial disease burden effectively.

HYPOTHESIS TESTING

1. **To analyze whether gender plays a role in the number of cancers cases in India.**

Dataset:

Zones	Males (X)	Females (Y)
Central Zone	4280917	3969935
East Zone	9270946	8637373
North East Zone	27541445	26998878
North Zone	32929570	29958509
South Zone	31641265	30410379
West Zone	46436043	42941495

Table 2.1: Representation of number of cancer cases among males and females in every zone

Hypothesis:

- H_0 (Null Hypothesis)
  There is no difference between the cancer cases of males and females
- H_1 (Alternative Hypothesis):
  There is a difference between the cancer cases of males and females

Calculations:

Using the table 2.1 we can find the following:

Degree of Freedom = (i-1) * (j-1) = (6-1) * (2-1) = 5

Where i is the number of rows

j is the number of columns

Observed Frequency:

Observed Frequency			
Zones	**Males (X)**	**Females (Y)**	**Total**
Central Zone	4280917	3969935	8250852
East Zone	9270946	8637373	17908319
North East Zone	27541445	26998878	54540323
North Zone	32929570	29958509	62888079
South Zone	31641265	30410379	62051644
West Zone	46436043	42941495	89377538
Total	152100186	142916569	295016755

Table 2.2: Observed Frequency of number of cancer cases among males and females in every zone

Expected Frequency:

Expected Frequency			
Zones	**Males (X)**	**Females (Y)**	**Total**
Central Zone	4253846.951	3997005.05	8250852
East Zone	9232894.758	8675424.24	17908319
North East Zone	28119058.09	26421264.9	54540323
North Zone	32422865.32	30465213.7	62888079

South Zone	31991629.07	30060014.9	62051644
West Zone	46079891.82	43297646.2	89377538
Total	152100186	142916569	295016755

Table 2.3: Expected Frequency of number of cancer cases among males and females in every zone

Final Calculation:

Class	Frequency		Observed - Expected	(Observed - Expected) ^2/Expected
	Observed	Expected		
Central Zone Male	4280917	4253846.95	27070.04863	172.2646681
Central Zone Female	3969935	3997005.05	-27070.04863	183.3341525
East Zone Male	9270946	9232894.76	38051.24171	156.8193978
East Zone Female	8637373	8675424.24	-38051.24171	166.8963909
North East Zone Male	27541445	28119058.1	-577613.0949	11865.15161
North East Zone Female	26998878	26421264.9	577613.0949	12627.5895
North Zone Male	32929570	32422865.3	506704.6848	7918.783091
North Zone Female	29958509	30465213.7	-506704.6848	8427.632916
South Zone Male	31641265	31991629.1	-350364.0653	3837.096823
South Zone Female	30410379	30060014.9	350364.0653	4083.663249
West Zone Male	46436043	46079891.8	356151.185	2752.690199
West Zone Female	42941495	43297646.2	-356151.185	2929.574185

Table 2.4: χ^2 calculation

The chi-square test statistics at **α=0.05** and **degree of freedom= 5** is given by:

$$\mathbf{X^2} = \sum \frac{(\boldsymbol{Observed\ Frequency - Expected\ Frequency})\text{^}\mathbf{2}}{\boldsymbol{Expected\ Frequency}}$$

X^2 calculated= 55121.49618

Using the χ^2 table, we get:

χ^2 tabulated= 11.07

Conclusion:

Since χ^2tabulated is less than X^2 calculated that is 11.07 < 55121.49618. Thus, we reject the null hypothesis (H0).

Hence, there is a difference between the cancer cases of males and females.

Regression Analysis:

Zones	Males (X)	Females (Y)
Central Zone	4280917	3969935
East Zone	9270946	8637373
North East Zone	27541445	26998878
North Zone	32929570	29958509
South Zone	31641265	30410379
West Zone	46436043	42941495

Table 2.5: Representation of number of cancer cases among males and females in every zone

Using the table 2.5, we calculate the mean for both Males (X) and Females (Y)

$$\bar{x} = \frac{\sum x}{n} = 25350031$$

$$\bar{y} = \frac{\sum y}{n} = 23819428.17$$

N = 6

Calculating the Covariance

The covariance between XX and YY is given by:

$$\text{Cov(X, Y)} = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{n-1} = 232099405607035$$

Calculating the Variance

The variance of X is given by:

$$\text{Var (X)} = \frac{\Sigma\,(x - \bar{x})^2}{n - 1} = 249779154652208$$

Calculating the Regression Coefficient

The formula for slope β_1 is:

$$\beta_1 = \frac{\text{Cov(X,Y)}}{\text{Var (X)}} = 0.93$$

The formula for intercept β_0 is:

$$\beta_0 = \bar{y} - \beta_1 \bar{x} = 263710.9756$$

The Regression Equation

The regression equation is given by:

$$Y = 263710.9756 + 0.93X$$

This means that every 1 increase in the male cancer cases there the cancer case of female increase by 0.93. This means the rise in cancer cases among female population is proportional to male population across all zones.

Correlation Analysis:

The correlation for the table 2.5 is given by:

Zones	Males (x)	Females (y)	x-x̄	y - ȳ	(x-x̄)*(y-ȳ)	(x-x̄)^2	(y-ȳ)^2
Central Zone	4280917	3969935	-21069114	-19849493	418211234370721	443907564744996	394002378973547
East Zone	9270946	8637373	-16079085	-15182055	244113555499523	258536974437225	230494799083710
North East Zone	27541445	26998878	2191414	3179449.8	6967490877064	4802295319396	10108901242683
North Zone	32929570	29958509	7579539	6139080.8	46531402600403	57449411452521	37688313478201
South Zone	31641265	30410379	6291234	6590950.8	41465213974995	39579625242756	43440632887417
West Zone	46436043	42941495	21086012	19122067	403208130712469	444619902064144	365653439978467

$$\text{Corr (r)} = \frac{\sum(x-\bar{x})*(y-\bar{y})}{\sqrt{(\sum(x-\bar{x})\text{^}2)*(\sum(y-\bar{y})\text{^}2)}} = 0.998596566$$

Conclusion:

The variable x and y are highly correlation. This means a high degree of correlation can be seen in the cancer cases of males and females and that is 0.9986.

2. **To assess whether geographical zone of residence influences the probability of individuals being diagnosed cancer positive.**

Dataset:

Zones	Total
Central Zone	41254260
East Zone	89541595
North East Zone	54540323
North Zone	157220197.5
South Zone	103419406.7

Table 2.6: Zonal representation of total cancer cases

Hypothesis

- H_0 (Null Hypothesis)
  Cancer cases are equally distributed across all zones
- H_1 (Alternative Hypothesis):
  Cancer cases are not equally distributed across all zones

Calculations:

Using the table 2.6 we can find the following:

Degree of Freedom = (i-1) = (6-1) = 5

Where i is the number of rows

Final Calculation:

Zones	Total	Expected	Chi Term
Central Zone	41254260	111569937.9	44315652.1
East Zone	89541595	111569937.9	4349270.95
North East Zone	54540323	111569937.9	29151015.4
North Zone	157220197.5	111569937.9	18678384.6
South Zone	103419406.7	111569937.9	595421.67
West Zone	223443845	111569937.9	112178705

Table 2.7: χ^2 calculation

The χ^2 term in table 2.7 is calculated using:

$$\chi^2 = \frac{(\textbf{Observed Frequency} - \textbf{Expected Frequency})\ \hat{}\,\mathbf{2}}{\boldsymbol{Expected\ Frequency}}$$

X^2 calculated = 209268449.2

Using the chi square table, we get

χ^2tabulated= 11.07

Conclusion:

Since χ^2tabulated is less than X^2 calculated that is 11.07 < 209268449.2. Thus, we reject the null hypothesis (H0). Hence, Cancer cases are not equally distributed across all zones

3. **To analyze the age group most at risk of developing cancer.**

Dataset:

Age Group	Total
00-04	23456241
05-09	24373027
10-14	25635098.5
15-19	26876491
20-24	30442587
25-29	29320648.5
30-34	25468697
35-39	23097636
40-44	19860908
45-49	17065502.5
50-54	13595898
55-59	10586263.5
60-64	8912146
65-69	5981238
70-74	4558818.5
75+	5785554.5

Table 2.8: Age wise distribution of cancer cases

Hypothesis:

- H_0 (Null Hypothesis)
  People of all age group are at the same risk of developing cancer
- H_1 (Alternative Hypothesis):
  People of all age group are not at the same risk of developing cancer

Calculations:

Age Group	Total	Expected	Chi Term
00-04	23456241	18438547.2	1365468
05-09	24373027	18438547.2	1910023
10-14	25635098.5	18438547.2	2808809
15-19	26876491	18438547.2	3861416
20-24	30442587	18438547.2	7814985
25-29	29320648.5	18438547.2	6422422
30-34	25468697	18438547.2	2680418
35-39	23097636	18438547.2	1177268
40-44	19860908	18438547.2	109721.8
45-49	17065502.5	18438547.2	102245.1
50-54	13595898	18438547.2	1271860
55-59	10586263.5	18438547.2	3343992
60-64	8912146	18438547.2	4921880
65-69	5981238	18438547.2	8416311
70-74	4558818.5	18438547.2	10448050
75+	5785554.5	18438547.2	8682800

Table 2.8: χ^2 calculation

The χ^2 term in table 2.8 is calculated using:

$$\chi^2 = \frac{(\textbf{Observed Frequency} - \textbf{Expected Frequency})\ \hat{}\,\mathbf{2}}{\boldsymbol{Expected\ Frequency}}$$

X^2 calculated = 65337669.4

Using the chi square table, we get

χ^2tabulated= 11.07

Conclusion:

Since χ^2tabulated is less than X^2 calculated that is 11.07 < 65337669.4. Thus, we reject the null hypothesis (H0). Hence, People of all age group are not at the same risk of developing cancer.

CONCLUSION

This study provides a detailed look at cancer incidence patterns across different regions, states, age groups, and genders in India. It uses secondary data from the National Centre for Disease Informatics and Research. The findings show a highly uneven distribution of cancer cases. Regions like the West, North, and South report significantly higher rates compared to the Central and East zones. These geographic differences point to factors like population density, healthcare infrastructure, diagnostic reach, and reporting methods affecting the cancer trends.

The analysis by gender shows a nearly equal distribution of cancer cases between males and females in most areas. This suggests that cancer is a common public health issue rather than one that affects just one gender. This balance highlights the need for gender-neutral approaches in cancer prevention, screening, and treatment, while still paying attention to specific risks for each gender. Age-related patterns show that cancer rates are particularly high among younger and middle-aged adults, especially in early adulthood, followed by a gradual decline in older groups. This underscores the importance of early screening, awareness campaigns, and lifestyle changes aimed at younger populations.

Comparisons between states and zones reveal significant differences in cancer burden. Metropolitan areas and healthcare hubs report higher numbers, likely due to better diagnostic access and patients coming

from nearby regions. Statistical analyses, such as chi-square tests and correlation studies, confirm strong links between demographic factors and cancer distribution, reinforcing that the patterns in the data are structured and not random.

However, the study has limitations due to its dependence on secondary data, incomplete district-level coverage, and potential reporting inconsistencies across states and regions. Still, the findings provide important evidence for policymakers, healthcare planners, and public health officials. Improving cancer surveillance systems, ensuring fair access to screening and treatment, and creating targeted interventions for specific areas will be crucial in lowering the cancer burden and enhancing health outcomes for the population. Overall, this study helps clarify cancer distribution patterns in India and serves as a foundation for future research and policy development.

LIMITATIONS

1. Limitations of the Study:

   The findings of the present study are based entirely on secondary data obtained from a government website, which inherently restricts the scope of the analysis. Only those states and regions that were available and reported on the official platform were included in the final dataset. As a result, certain states that fall within specific zones may not have been represented due to the unavailability or incomplete publication of their data. This limitation may affect the completeness of zonal comparisons and could lead to an underestimation or overestimation of cancer burden in some regions.

   Additionally, the study relies on aggregate data, which limits the ability to explore individual level variation or micro level determinants of cancer incidence. Consequently, the finding primarily reflects broad regional patterns rather than localized or personalized trends.

2. Limitations of the Research Procedure:

   The research employed statistical techniques such as chi-square, correlation, covariance and linear regression which are appropriate for examining associations and linear relationships. However, these methods assume data accuracy and uniform reporting practices across regions. Variations in data collection procedures, reporting standards, and diagnostic coverage across states and zones may influence the observed patterns and introduce reporting bias.

Furthermore, the absence of additional explanatory variables – such as socioeconomic status, environmental exposure, healthcare accessibility and lifestyle factors – limits the ability to establish causal relationships. The analysis therefore focuses on statistical association rather than causation, which may restrict deeper interpretation of underlying factors influencing cancer distribution.

3. Limitations Related to Data Coverage and Representation:
   Another important limitation of this study is the uneven representation of districts and states within the dataset. Not all districts within the included states were present in the final data, and some states have a significantly higher number of district compared to others. This imbalance may affect the proportional representation of cancer cases, particularly in states with more extensive administrative divisions.

   Moreover, differences in the number of reporting cancer registries and diagnostic centers across regions can result in unequal case capture. States or districts with better reporting infrastructure may appear to have a higher cancer burden, not necessarily due to higher incidence, but due to more effective detection and documentation systems.

4. Generalizability of Findings:
   Due to the reliance on officially reported data and the exclusion of non-reporting regions, the findings of this study may not be fully generalized to the entire population. The results should therefore

be interpreted with caution, especially when drawing comparisons across zones or making policy recommendations. Despite these limitations, the study provides valuable insights into broad cancer distribution pattern and serves as a foundation for more detailed future research incorporating primary data and wider regional coverage.

SUGGESTIONS & RECOMMENDATIONS

1. **Strengthening Government Role in Cancer Control**

Improved Data Reporting and Surveillance:

Government agencies should improve cancer surveillance systems by ensuring consistent data collection and timely reporting across all states and districts. Expanding cancer registries will help create a clearer picture of cancer incidence in different areas.

Policy Support and Funding:

The government should increase funding for cancer prevention, early detection programs, and treatment facilities, especially in high-burden areas such as the North, South, and West regions.

National Awareness Initiatives:

Nationwide campaigns should educate people about cancer risk factors, early symptoms, and the importance of timely screening, particularly among vulnerable age groups.

2. **Enhancing Healthcare Infrastructure and Accessibility**

Expansion of Screening and Diagnostic Services:

Screening facilities should be expanded, especially in rural and underserved areas, to ensure early detection and reduce differences in reported cancer cases across regions.

Strengthening Treatment Capacity:

High-burden regions should have more oncology centers, better diagnostic tools, and trained healthcare professionals to effectively manage the increasing number of patients.

Affordable Cancer Care:

Financial assistance programs and insurance coverage should be improved to lessen the economic burden of cancer treatment and ensure fair access for all income groups.

3**. Gender- and Age-Specific Interventions**

Targeted Screening Programs:

Since cancer affects both males and females nearly equally, gender-neutral screening programs should be promoted. Efforts should also address specific gender-related cancers through targeted initiatives.

Early-Age Interventions:

Considering the higher incidence among younger and middle-aged populations, early screening, lifestyle changes, and awareness programs in schools should be prioritized.

4. **Research and Capacity Building**

Encouraging Further Research:

Future studies should include primary data, district-level analysis, and other factors like socioeconomic status, environmental exposure, and lifestyle choices to offer better insights into cancer trends.

Training and Skill Development:

Ongoing training programs should be provided for healthcare professionals to enhance early diagnosis, reporting accuracy, and patient care outcomes.

5. **Community Engagement and Support Systems**

Community-Based Outreach Programs:

Local awareness campaigns and mobile screening units can help reach people in remote areas and increase participation in early detection efforts.

Psychological and Social Support:

Counseling services and patient support groups should be included in cancer care systems to help address the emotional and psychological challenges faced by patients and their families.

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